

Abhijeet Melkani

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Education

M.S. & Ph.D. in Physics Sep 2019 – Sep 2024
University of Oregon, Eugene, USA
Advisor: Jayson Paulose
Thesis: [Non-Hermitian Structures in Soft Matter](#)

B. Tech. (with Honours) in Engineering Physics Jul 2015 – May 2019
Indian Institute of Technology Bombay, Mumbai, India

Research Interests

My research lies at the interface of active matter, biophysics, and dynamical metamaterials. My work combines analytical approaches—including hydrodynamics, kinetic theory, and tools from non-Hermitian and topological dynamics—with data-driven techniques such as machine learning and agent-based simulations.

Publications & Preprints

- *The fluctuating shape of fish schools: a spectroscopic probe of their collective motion*
A. Melkani, A. Delory, M. Munari, G. Calvet, D. Bartolo (in preparation)
- *Space-time Floquet operator: Non-reciprocity and fractional topology of space-time crystals*
A. Melkani, J. Paulose, [Arxiv.2510.16562](#) (2025)
- *Space-time symmetry and nonreciprocal parametric resonance in mechanical systems*
A. Melkani, J. Paulose, [Phys. Rev. E 110, 015003](#) (2024) [Editors' suggestion]
- *Degeneracies and symmetry breaking in pseudo-Hermitian matrices*
A. Melkani, [Phys. Rev. Res. 5, 023035](#) (2023)
- *Delocalization of interacting directed polymers on a periodic substrate: Localization length and critical exponents from non-Hermitian spectra*
A. Melkani*, A. Patapoff*, J. Paulose, [Phys. Rev. E 107, 014501](#) (2023)
- *Electron vortex beams in nonuniform magnetic fields*
A. Melkani, S. J. van Enk, [Phys. Rev. Res. 3, 033060](#) (2021)
- *Eigenstate extraction with neural-network tomography*
A. Melkani, C. Gneiting, F. Nori, [Phys. Rev. A 102, 022412](#) (2020) [Editors' suggestion]

*Joint first authors

Scholarships & Awards

American Physical Society Irwin Oppenheim Award	2026
UO Division of Global Engagement Endowed Scholarship	2023
RIKEN Visiting Researcher Award and Travel Award	2018
Outstanding Engineering Student Award sponsored by Uttarakhand Technical University	2016
Ranked 311 in the Indian National Physics Olympiad (Certificate of Excellence)	2015
Ranked 720 in the Indian National Chemistry Olympiad (Certificate of Excellence)	2015
Kishore Vaigyanik Protsahan Yojana (KVPY) fellowship (National Rank 368)	2015